Q1 A coil, having an inductance of $0.2\ \mathrm{H}$ and across a $100\ \mathrm{V}$ a.c. supply. Calculate the current when the frequency is:

a) $30\ \mathrm{Hz}$
b) $50\ \mathrm{Hz}$
c) $500\ \mathrm{Hz}$

Q2 A coil of inductance $0.1\ \mathrm{H}$ and negligible resistance is connected in series with a $25\ \Omega$ resistor. The circuit is energized from a $230\ \mathrm{V}$, $50\ \mathrm{Hz}$ source. Calculate:

a) The current in the circuit
b) The potential difference across the coil
c) The potential difference across the resistor
d) The phase angle of the circuit

Draw to scale a phasor diagram representing the current and the component voltages.

Q3 A $15\ \Omega$ non-reactive resistor is connected in series with a coil of inductance $0.08\ \mathrm{H}$ and negligible resistance. The combined circuit is connected to a $240\ \mathrm{V}$, $50\ \mathrm{Hz}$ supply. Calculate:

a) The reactance of the coil
b) The impedance of the circuit
c) The current in the circuit

Q4 If voltage $v(t) = 25\sin(100t - 15^{\circ})\ \mathrm{V}$ is applied to a $50\ \mu\mathrm{F}$ capacitor, derive an expression for the instantaneous current through the capacitor.

Q5 If the current flowing through a series connected $680\ \Omega$ resistor and a $10\ \mu\mathrm{F}$ capacitor is $i(t) = 55\sin(100t + 45^{\circ})\ \mathrm{mA}$, derive an expression for the instantaneous voltage of the supply.

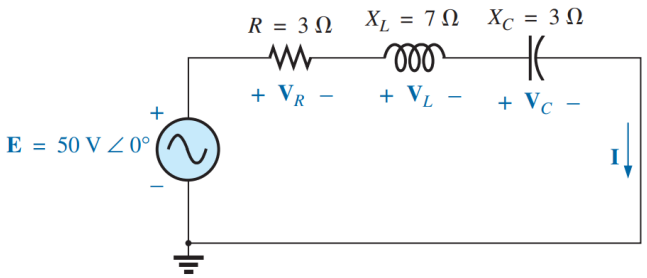
continues…

Q6 For the following three voltages,

$$e_1 = 80\cos(\omega t)$$
$$e_2 = 60\sin(\omega t)$$
$$e_3 = 100\cos\left(\omega t - \frac{\pi}{3}\right)$$

(a) Convert these voltages into complex phasor form.
(b) Calculate the sum of these voltages using complex arithmetic.
(c) Express the sum of these voltages in form of $E_m \sin(\omega t \pm \phi)$.
(d) When this resultant voltage is applied to a circuit consisting of a $10\ \Omega$ resistor and a capacitor of $17.3\ \Omega$ reactance connected in series, find an expression for the instantaneous current, expressed in the same form.
(e) Draw the phasors representing all the quantities.

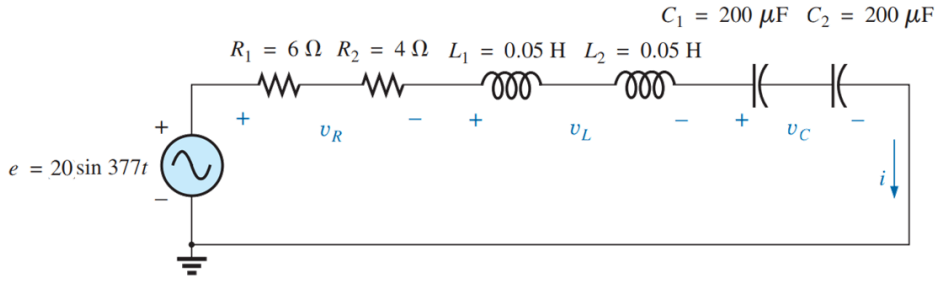
Q7 For the series RLC circuit below



Write an expression for,

a) The total impedance (complex phasor) of the of the series RLC circuit.
b) The instantaneous current, I.
c) The instantaneous voltage for V_R, V_L, and V_C.

Q8 For the circuit below,



a) Derive the instantaneous $i(t)$, $v_R(t)$, $v_L(t)$, and $v_C(t)$.
b) Draw the phasor diagram.

End